LINK TO TINKERCARD :

https://www.tinkercad.com/things/9gCOjN7lWGd-epic-jaban?sharecode=fFZdjwHzE_Z2znY5p0K-mLgLOR3_3UVkw607qz67UE4

```
const int sensorPin = A0;
const int ledPin = 7;

float voltage;
float current; float
power;
float energy = 0.0;

unsigned long previousTime = 0;

void setup() {
pinMode(ledPin, OUTPUT);
  Serial.begin(9600);

  Serial.println("IIoT Energy Monitoring System");
}

void loop() {    int sensorValue =
analogRead(sensorPin);

  // Simulated voltage and current values
voltage = 230.0;   current = (sensorValue
/ 1023.0) * 10.0;

  // Calculate power in watts
  power = voltage * current;

  // Calculate energy in kWh
  unsigned long currentTime = millis();

  if (previousTime != 0) {
float elapsedHours =
     (currentTime - previousTime) / 3600000.0;

    energy += (power * elapsedHours) / 1000.0;
  }

  previousTime = currentTime;

  Serial.print("Voltage: ");
  Serial.print(voltage);
```

```
  Serial.println(" V");

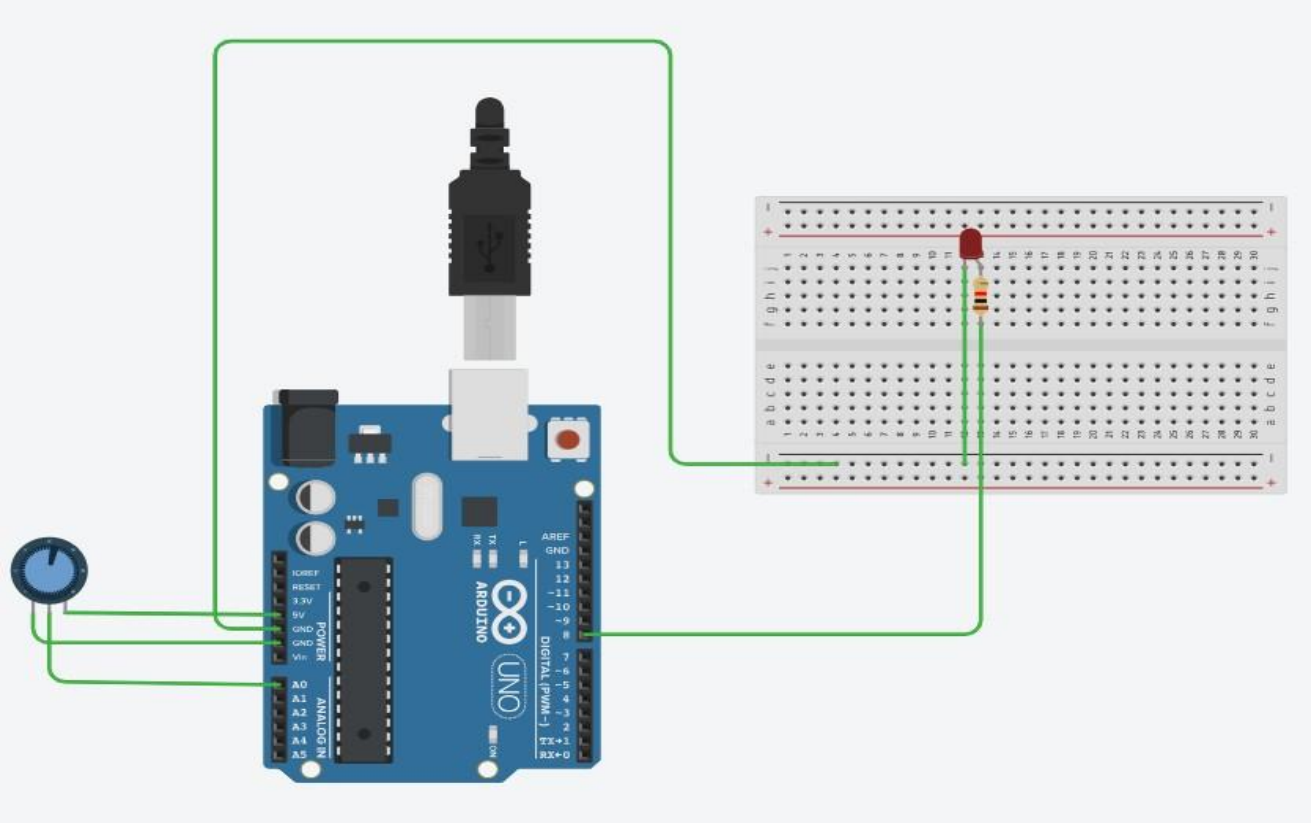
  Serial.print("Current: ");
  Serial.print(current, 2);
  Serial.println(" A");

  Serial.print("Power: ");
  Serial.print(power, 2);
  Serial.println(" W");

  Serial.print("Energy: ");
  Serial.print(energy, 6);
  Serial.println(" kWh");

  // Optimization alert
if (power > 1500) {
    digitalWrite(ledPin, HIGH);
    Serial.println("ALERT: High energy consumption!");
    Serial.println("Optimization: Reduce unnecessary loads.");
  } else {
    digitalWrite(ledPin, LOW);
    Serial.println("Energy consumption is normal.");
  }

  Serial.println("-----------------------");
  delay(1000);
}
```


IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
A0
A1
A2
A3
A4
A5
ANALOG IN
TX
RX
L
ON
ARDUINO
UNO
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)